

Replay inputs

Training episodes only

Original source episode



$s_0, A_{\text{ref}}, T = 400$

Reset to the same initial state

Bounded pulse from same-task errors



$P \in \mathbb{R}^{8 \times 14}$

σ : transfer 0.03, insert 0.08

Twelve endpoints per source episode

Successful recovery

Same pulse, ordered scale backtracking



Endpoint u (seeded)

①

Candidate scale



$\lambda = 1, \frac{1}{2}, \frac{1}{4}, 0$
 $\lambda = 0$: identity fallback

②

Full episode replay



400 steps at 50 Hz
ref. $\rightarrow$ pulse $\rightarrow$ ref.

③

Terminal acceptance



$= \frac{1}{4}$ Accept if $r_{399} = 4$

pass

④

Recovery action labels



$Y_t = A_{\text{exec}, t:t+16}$

$t \geq u$

⑤

Recovery window extraction



$(\tilde{o}_t, Y_t), t \in U(u)$

Y_t

Export supervision

Verify, trace, train

Check executed recovery labels

Y_t



0

$Y_t - A_{\text{ref}, t:t+16} = 0$

A_{ref}

Write $\{\tilde{o}_t, Y_t, \text{metadata}\}$

Traceable recovery record



Y_t | executed reference chunk

$\tilde{o}_t$ | current replay observation

λ | accepted scale and replay hash

task · source · endpoint · start · seed



Original data D_0

Recovery windows

1:1 training mixture

Action-chunk policy



Input o_t

Y_t

MSE target

Predict 16 steps
Execute 8; observe